

Lecture no. 7:-

Quantify Chance Entropy Information, Motivation

Quantifying Chances:-

This is a thinking lesson!!

Problem:-

- A disease is prevalent in 0.2% of a population.
- We have a test that, given to a sick person, gives a the positive result 85% of the time
- of all the people ever tested, 8% were positive.

Q: If Ali is tested and test comeback positive, What are the chances that she actually has the disease?

☐ 85% ☐ 77% ☐ 21% ☒ 2%

Q: What is "information"?

"Gooblede coope gogle" less information
"I can speak Gibberish!" more information

"quantification"

Let's say you're doing cryptanalysis

Key 1 $\rightarrow$ "Gooblede oompa gogie"

$\rightarrow 29$

Key 2 $\rightarrow$ "I can speak Gibberish"

$\rightarrow 1058$

} Entropy

$\rightarrow$ basic foundation of probability

consider as
it is quantification unit

We can't eyeball 2.4 billion decryptions!

Chance $\rightarrow$ Certainty / Uncertainty $\rightarrow$ Entropy

— Video streaming — file size

— Password strength

— Huffman's codes — compression

— Computer vision

— Machine learning — throw away data / information

— Bits — foundation of all computation.

(consider bit unit of entropy)

(Use whole ~~compression~~ ~~compression~~)

All of above are applications of Entropy

Quantifying Chances (The lecture 8 do study on step 2r Certainty / Uncertainty.)